

Training Configuration and Summary:

Sequencing:

- timesteps: 60
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: gru_hyperparameter_tuning

GRU Training Report

Report Generated: 2025-04-09_11-48-54

Training Start: 2025-04-09 11:34:34

Training End: 2025-04-09 11:48:40

Training Duration: 846.02 seconds

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 192 units, Layer 2: 256 units

Performance Metrics:

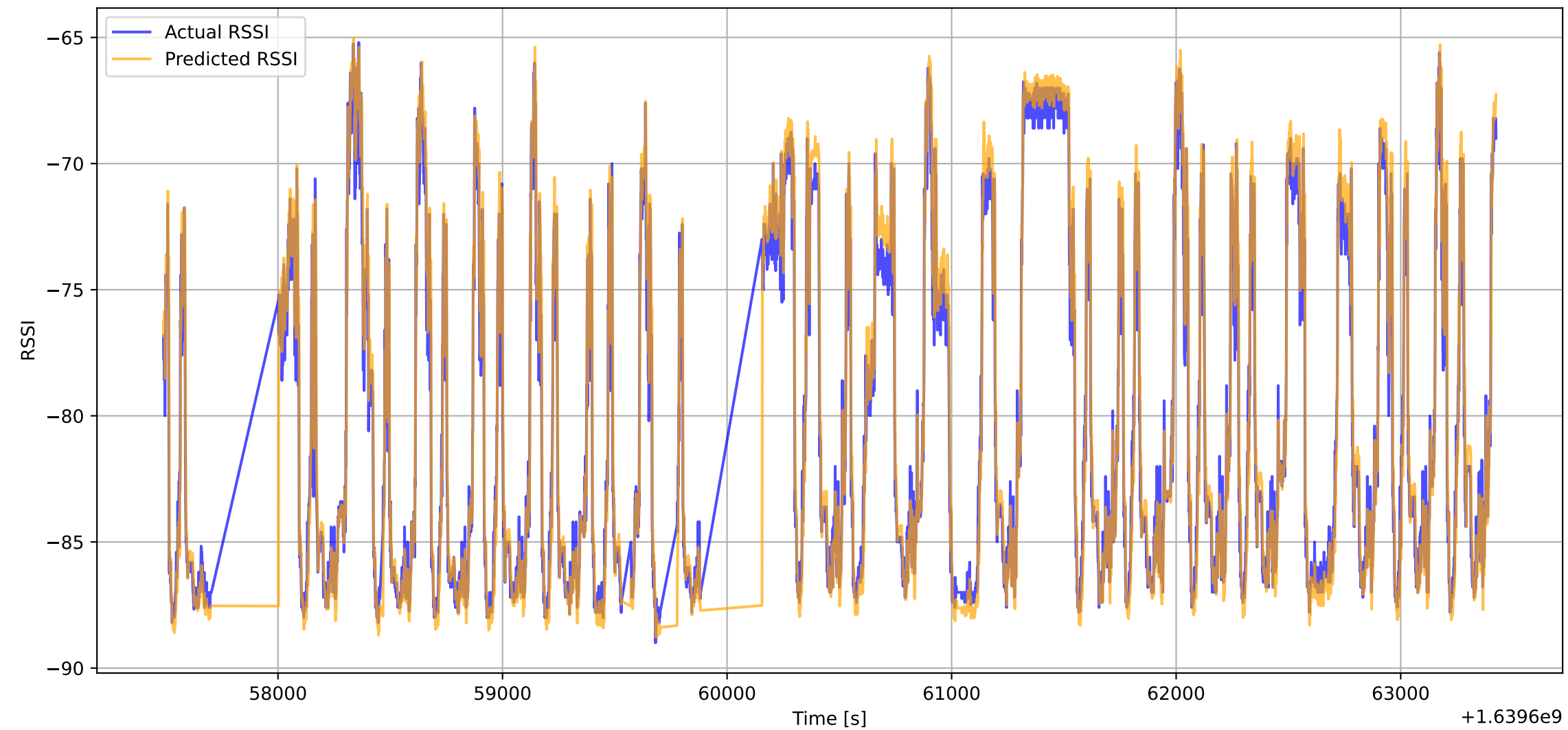
MAE: 1.0607

MSE: 2.2807

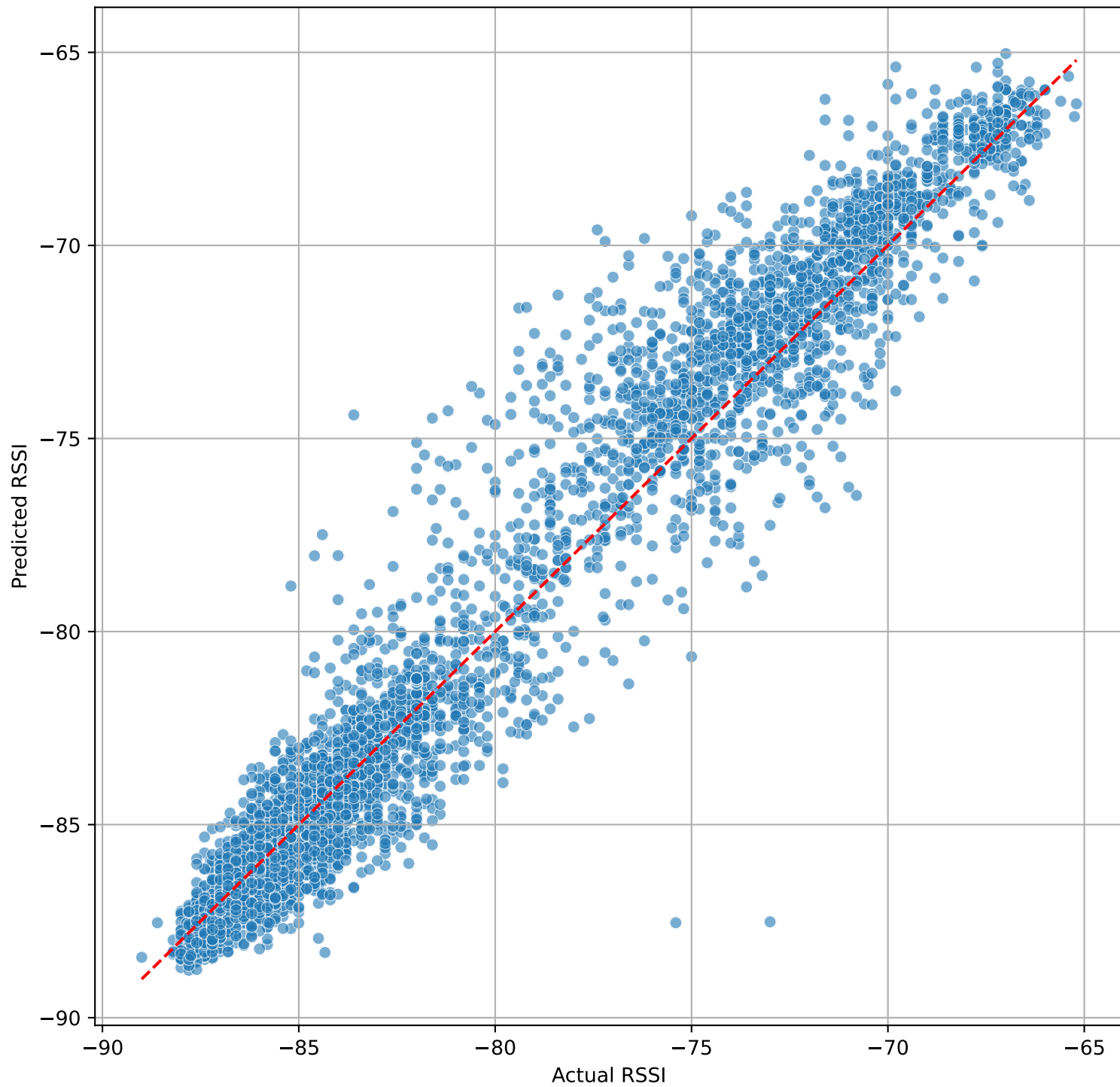
RMSE: 1.5102

R2 Score: 0.9482

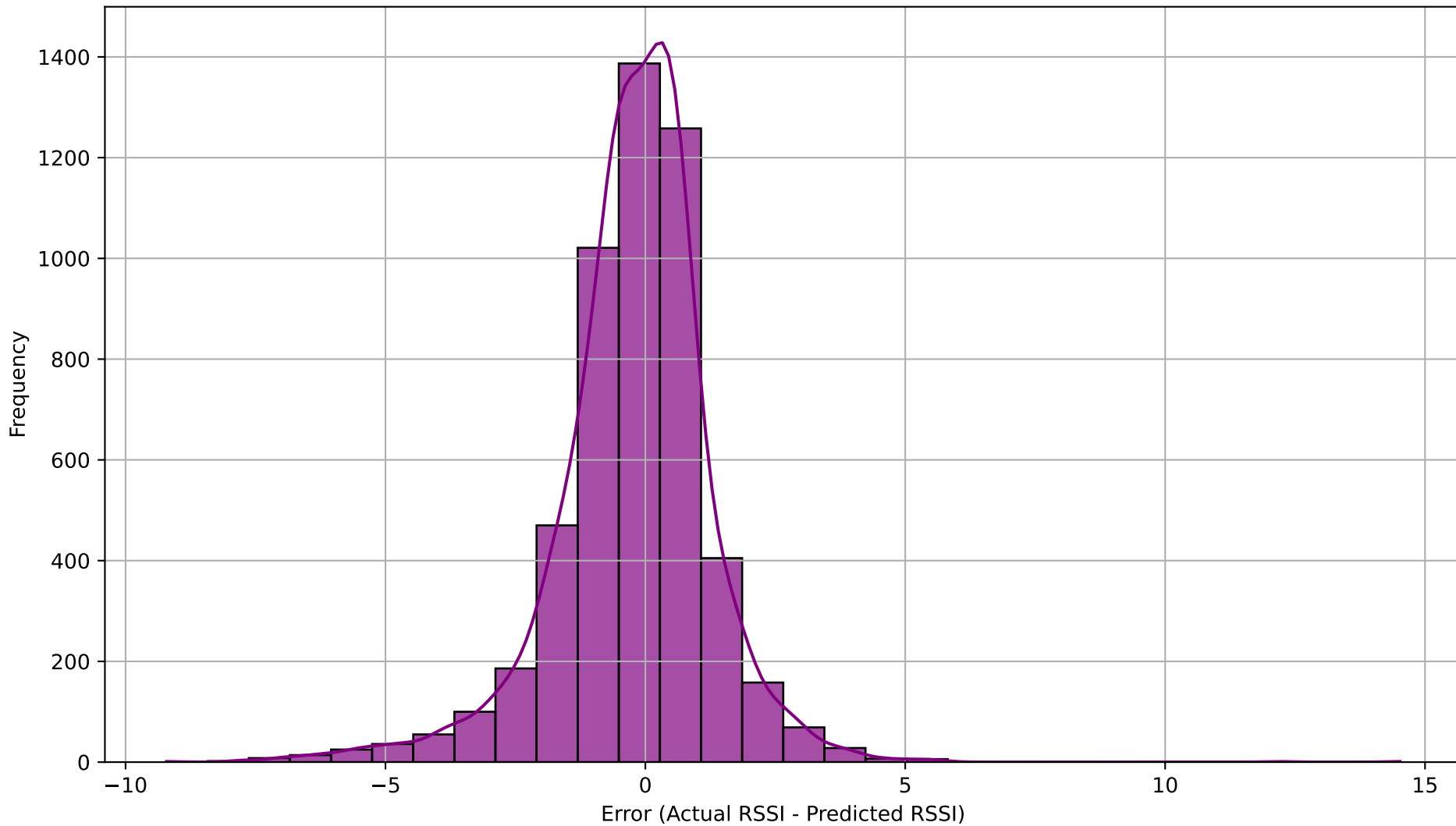
Actual vs Predicted RSSI Over Time



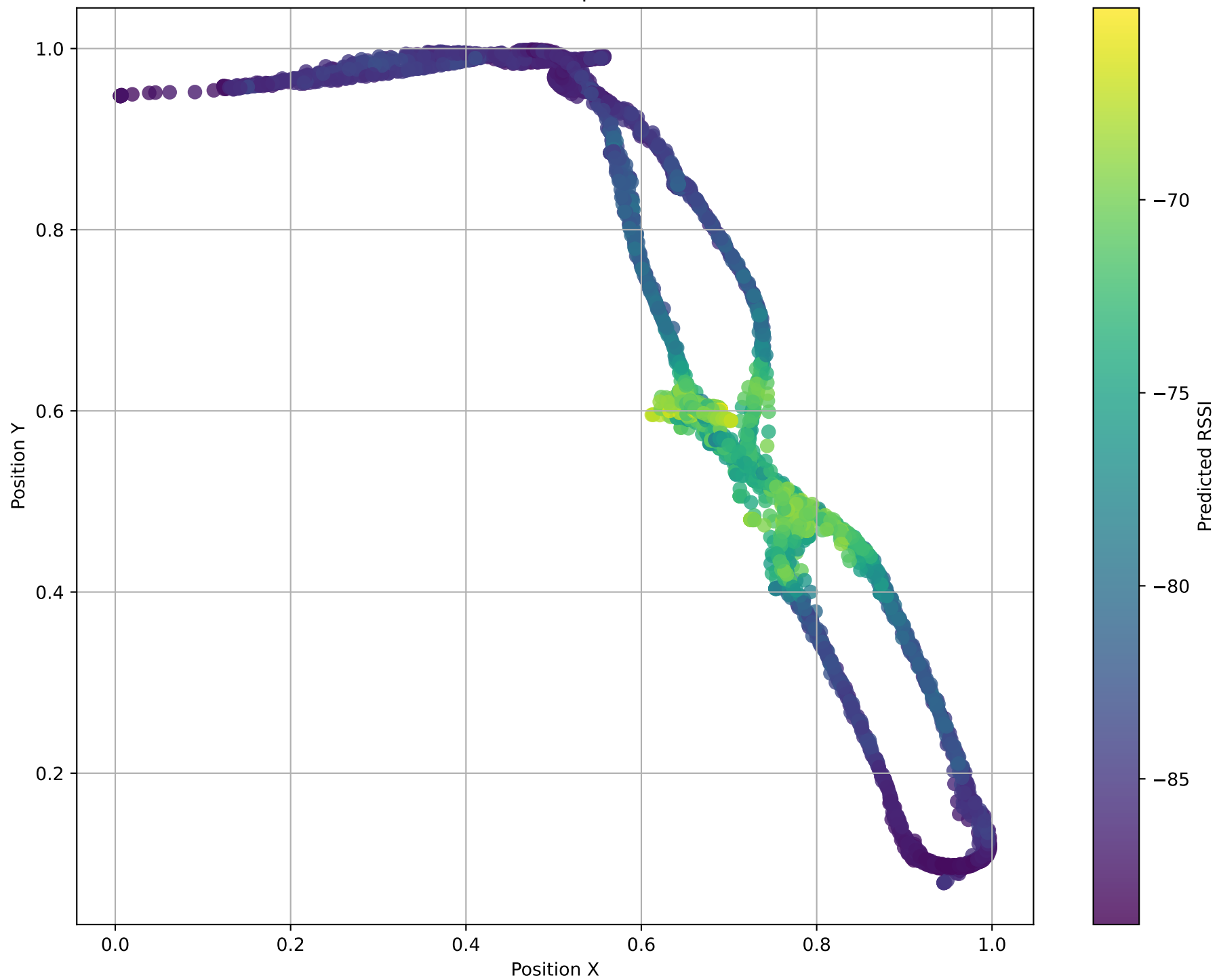
Actual vs Predicted RSSI



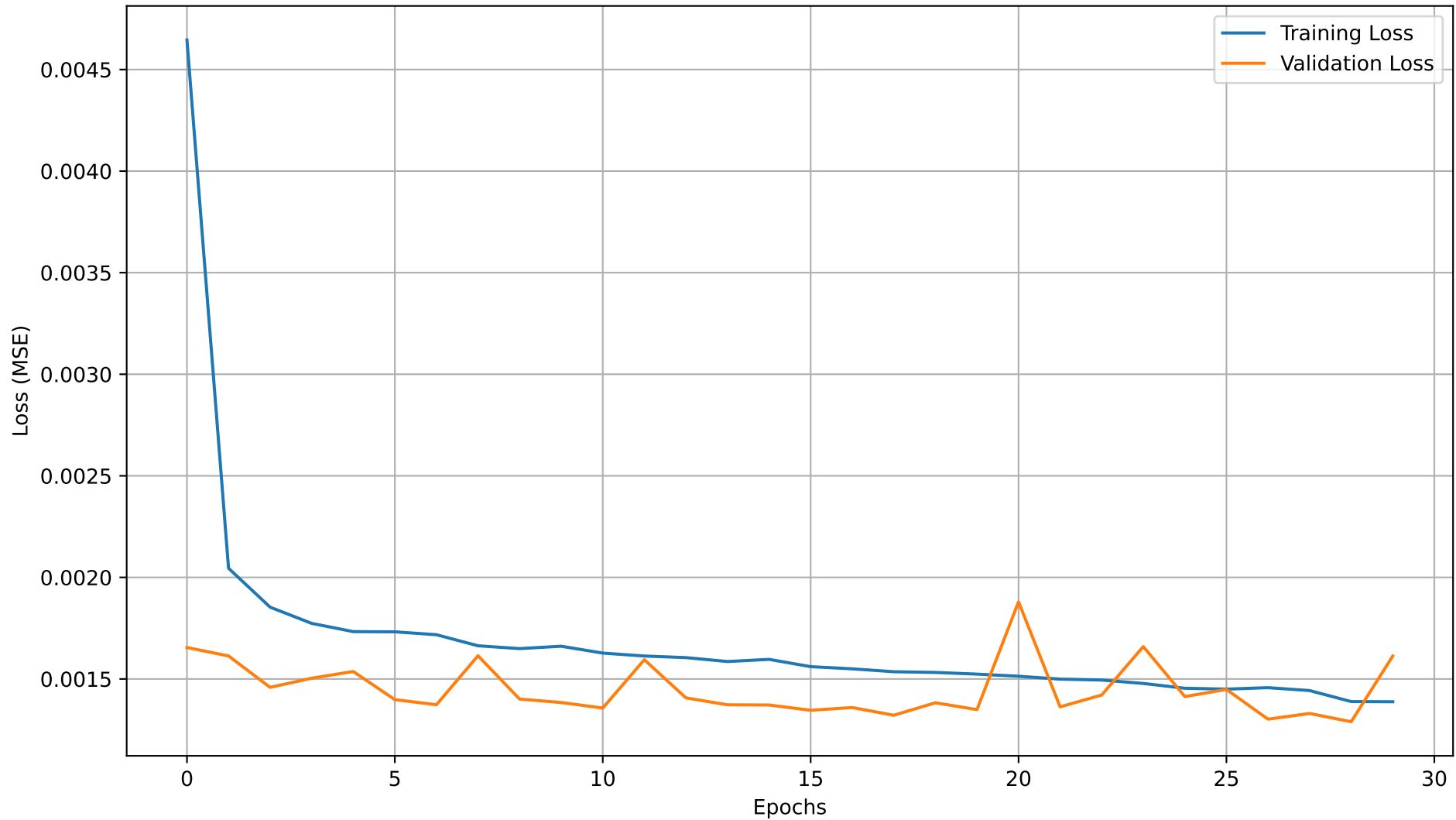
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

